How the Damper Pressure Drop Estimator Calculates Pressure Drop

Method description with worked examples, for fire/smoke dampers in IP units

This document explains, step by step, how the program turns a damper size and an airflow into a pressure drop: where every number comes from, which equations are used, and how test data, measured free areas, the blade table and the multi-section size rules feed into the result. Three worked examples at the end follow a tested size, a size next to a tested one, and a multi-section damper through every step.

Sample data. The worked examples use the SAMPLE-FSD-100 model shipped with the program, whose data is made up for demonstration. The method is the same for real models; only the numbers change. Every number in this document is calculated by the program itself (docs/make_method_pdf.py).

1. The idea in one equation

A damper is an obstruction in the duct. The frame, sleeve, blades and linkage block part of the opening, so the air has to speed up to get through the **free area** that is left. Like any obstruction, the pressure it costs is proportional to the **velocity pressure** of that faster air. The program therefore uses:

$$\Delta P = C_{free} \times VP_{free}$$

$$VP_{free} = (\rho / 0.075) \times (V_{free} / 4005)^2 \qquad V_{free} = Q / A_{free}$$

ΔP is the pressure drop (in. wg), Q the airflow (cfm), A_{free} the free area (ft²), ρ the air density (lb/ft³) and C_{free} a dimensionless **loss coefficient**. Everything except C_{free} follows from the size, the geometry and the air conditions. C_{free} is learned from your test data. Because the pressure drop scales with velocity squared, doubling the airflow quadruples the pressure drop.

The six steps

Step	What happens	Uses
1	Check the size against the model's limits and split it into sections if needed	size_rules.json
2	Find the free area	free_area_data.csv, geometry.json, blade_table.csv
3	Work out the face and free-area velocities	airflow or face velocity
4	Find the air density and the velocity pressure	temperature, altitude
5	Choose the loss coefficient C_{free}	damper_data.csv (test data), size_model rule
6	Multiply: $\Delta P = C_{free} \times VP_{free}$	

2. Step 1: size check and section layout

The size is first checked against the model's minimum and maximum overall width and height in **size_rules.json**. A size outside the limits is rejected with a message rather than estimated, because the model is not made in that size.

If the model has a maximum section width or height, a larger damper is split into the fewest **equal** sections that keep each one within the limit:

$$n_w = \lceil W / W_{max\ section} \rceil \qquad n_h = \lceil H / H_{max\ section} \rceil$$
$$section\ size\ \ W_s = W / n_w,\ \ H_s = H / n_h$$

⌈ ⌉ means round up to the next whole number. With a 48 in maximum section width, a 96 in wide damper becomes 2 sections of 48 in, and a 100 in wide damper becomes 3 sections of 33.33 in. Each section is treated as its own damper with its own frame, blades and other blockage. With no section limits, every damper is a single section ($n_w = n_h = 1$).

3. Step 2: free area

The free area of the requested size is taken from the first of these sources that has it:

Priority	Source	When it applies
1	free_area_ft2 column in damper_data.csv	The size has a free area entered with its test data
2	Measured free-area table, free_area_data.csv	The size is listed in the table (within 0.01 in)
3	Geometry estimate	Any other size
(override)	Free area typed into the Estimate tab	Replaces all of the above for that estimate

Geometry estimate

For each section, the frame obstruction f is removed from every side to give the clear opening, then the blades (each blocking a strip of thickness t across the clear width) and the other blockage O (linkage, jackshaft, stops) are subtracted:

$$w_{open} = W_s - 2f_w \qquad h_{open} = H_s - 2f_h$$
$$A_{section} = w_{open} \times h_{open} - N \times t \times w_{open} - O$$
$$A_{free} = n_w n_h \times A_{section} - A_{joints} \qquad (in^2,\ divide\ by\ 144\ for\ ft^2)$$

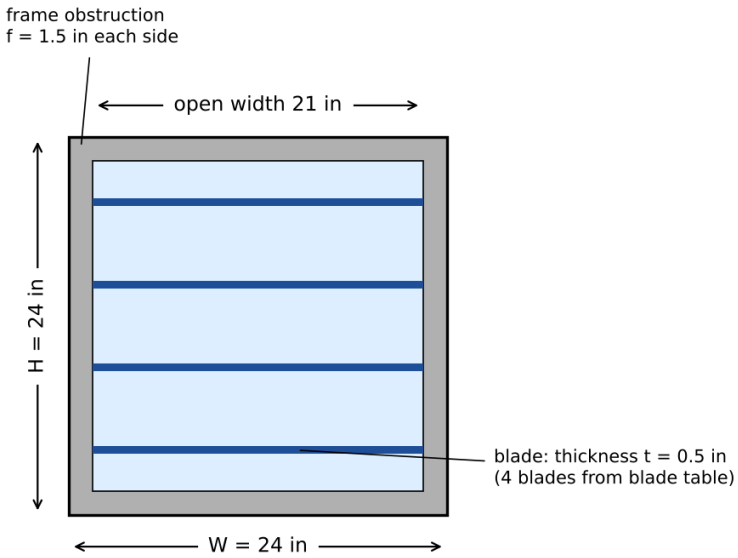


Figure 1. Geometry of a single 24 x 24 section (SAMPLE-FSD-100 geometry). Free area = 21 × 21 − 4 × 0.5 × 21 − 0 = 399 in² = 2.771 ft².

Number of blades *N*

N is looked up in the **blade table** (blade_table.csv) by the **nominal section height** H_s, the way blade schedules are usually written. Each row covers a height range including both ends; a height on a boundary shared by two rows uses the lower row. If the height falls in a gap between two rows, the program stops with an error, because the table is probably missing a row. If the height is above or below the whole table, or the table is empty, the blade pitch *p* is used instead (with a warning on the estimate when a table exists). The pitch rule works on the **open height**, because blades only fill the clear opening:

$$N = \lceil h_{open} / p \rceil \quad \textit{(only when the blade table does not cover the section height)}$$

Joints between sections

Each section already has its own frame, so back-to-back frames at a joint are counted automatically. The optional **joint blockage** *j* adds extra width blocked along every joint (a mullion or connecting angles). There are $n_w - 1$ vertical joints of length H and $n_h - 1$ horizontal joints of length W; where they cross, the overlap would be counted twice, so it is taken off once:

$$A_{joints} = j \times [\,(n_w - 1) \times H + (n_h - 1) \times W\,] - j^2 \times (n_w - 1)(n_h - 1)$$

Calibrating the geometry

When measured free areas are available, **Calibrate geometry** finds the frame obstruction, blade thickness and other blockage that make the geometry estimate match them best (least squares). The frame obstruction is searched in 1/32 in steps from 0 to 6 in. For each frame value, the equations above are linear in t and O, so those two follow directly from a linear least-squares fit, constrained to be zero or positive. The blade counts come from the blade table (or the pitch, which is not fitted) and multi-section sizes are split by the size rules, exactly as for an estimate. Sizes listed in the measured table always use their measured value; the calibrated geometry only affects sizes that are not listed.

Calibrate before relying on untested sizes. A tested size's own estimate does not depend on its free area: its coefficient was fitted with the same free area, so the two cancel. Free area only matters for sizes that were not tested, through the free-area ratio in the size model. If the geometry estimate disagrees with the measured table, a size just next to a tested one can get a noticeably different pressure drop, because it takes its free area from the geometry. The SAMPLE-FSD-100 geometry is left uncalibrated (it over-estimates the 24 x 24 free area by 6.2%).

4. Step 3: velocities

The airflow can be entered directly or as a face velocity; the other follows from the face area:

$$A_{face} = W \times H / 144 \ \ (ft^2) \qquad Q = V_{face} \times A_{face}$$

$$V_{free} = Q / A_{free} = V_{face} / R \qquad \text{free-area ratio } \ R = A_{free} / A_{face}$$

R is the fraction of the nominal opening that is actually open. A damper with R = 65% makes the air move 1 / 0.65 = 1.54 times faster than the face velocity.

5. Step 4: air density and velocity pressure

Air density is found from the temperature T (°F) and altitude z (ft) using the standard-atmosphere pressure (ASHRAE) and the ideal-gas law, referenced to standard air (0.075 lb/ft³ at 70 °F and sea level). Humidity is not included; its effect is usually under 1%.

$$\rho = 0.075 \times (1 - 6.8754 \times 10^{-6} \times z)^{5.2559} \times 529.67 / (T + 459.67)$$

The velocity pressure is the pressure equivalent of the air's speed. For standard air, V = 4005 × √VP (fpm, in. wg), which gives:

$$VP_{free} = (\rho / 0.075) \times (V_{free} / 4005)^2 \quad (in.\ wg)$$

The factor ρ / 0.075 corrects for non-standard air: lighter air (hot, or at altitude) carries less velocity pressure at the same velocity, so it causes a smaller pressure drop. Test data is assumed to be at standard air, which is how AMCA 500-D test reports present it.

6. Step 5: the loss coefficient C_{free}

5a. Fitting each tested size

For every size in damper_data.csv, each test point's face velocity is converted to a free-area velocity pressure $x_i = VP_{free,i}$ (standard air), and paired with its measured pressure drop y_i. The model says $y = C \times x$, a straight line through zero (no flow, no pressure drop). The best-fitting slope by least squares is:

$$C_{free} = \Sigma\, x_i y_i \,/\, \Sigma\, x_i^2 \qquad R^2 = 1 - \Sigma(y_i - Cx_i)^2 \,/\, \Sigma(y_i - \bar{y})^2$$

R^2 (shown in the program for every size) measures how well the points follow the velocity-squared law: 1.000 is a perfect fit. Because the fit minimises absolute errors, it is steered mostly by the high-velocity points, where the pressure drops are largest. Two further checks show whether the lower velocities agree (5b).

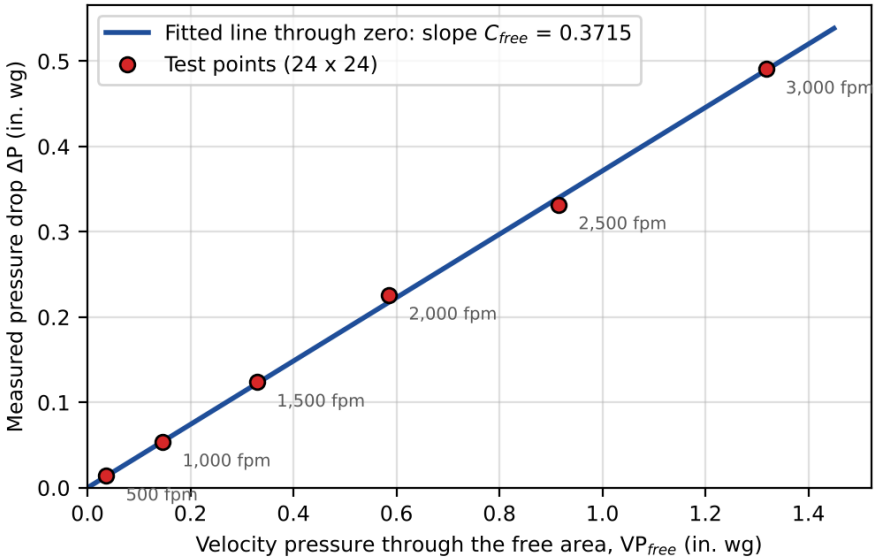


Figure 2. Fitting the 24 x 24 size of SAMPLE-FSD-100. Plotted against the free-area velocity pressure, the test points fall on a straight line through zero; its slope is C_{free} = 0.3715 (R^2 = 0.999).

5b. Fit diagnostics: exponent and residuals

For each tested size the program also fits $\Delta P = k \times V^n$ on log scales. For turbulent flow through a damper the exponent *n* should be close to 2. A size whose *n* is outside 2 ± 0.2 is marked in the program, and an estimate that uses it carries a warning, because the velocity-squared fit then hides a problem with the data. The **residual** of each test point, (measured − fitted) / fitted, shows where the fit misses; both are listed in the Test data & model tab and by --list on the command line.

Size	n	Largest residual	Size	n	Largest residual
12 x 12	1.98	+3.3%	18 x 18	2.00	+3.1%
24 x 24	1.99	+3.9%	36 x 24	2.02	-4.5%
36 x 36	2.01	-7.3%	48 x 48	2.02	+4.2%

5c. The size model

C_{free} differs between sizes. Small dampers lose a larger share of their opening to the frame and blades (lower R), and the flow through them contracts and expands more sharply, so they have a higher coefficient. To estimate sizes that were not tested, the program fits the coefficients of all tested sizes against their free-area ratio, in one of two shapes chosen per model by the **size_model** rule (size_rules.json, the Size rules sheet, or the Sizes & blades tab):

$$\textit{linear (default):}\quad C_{free}(R) = a + b \times R$$

$$\textit{curved:}\quad C_{free}(R) = a + b \times (1/R - 1)^2$$

The curved shape is the shape of the classic loss coefficient of a grille or perforated plate (Idelchik): it rises steeply as R gets small and levels off as R approaches 1. A straight line can fit well inside the tested range but under-predicts small dampers below it, and keeps falling above it. Both shapes are fitted every time and both are plotted in the Test data & model tab, so you can see which one follows your data before choosing. For SAMPLE-FSD-100, whose made-up data is nearly linear:

Shape	Fitted size model	R^2 across sizes
linear (in use)	$C_{free} = 0.8799 - 0.7747 \times R$	0.998
curved	$C_{free} = 0.3138 + 0.1329 \times (1/R - 1)^2$	0.868

This needs at least 3 tested sizes whose ratios differ by at least 2 percentage points. With fewer, the average coefficient of the tested sizes is used for every size instead.

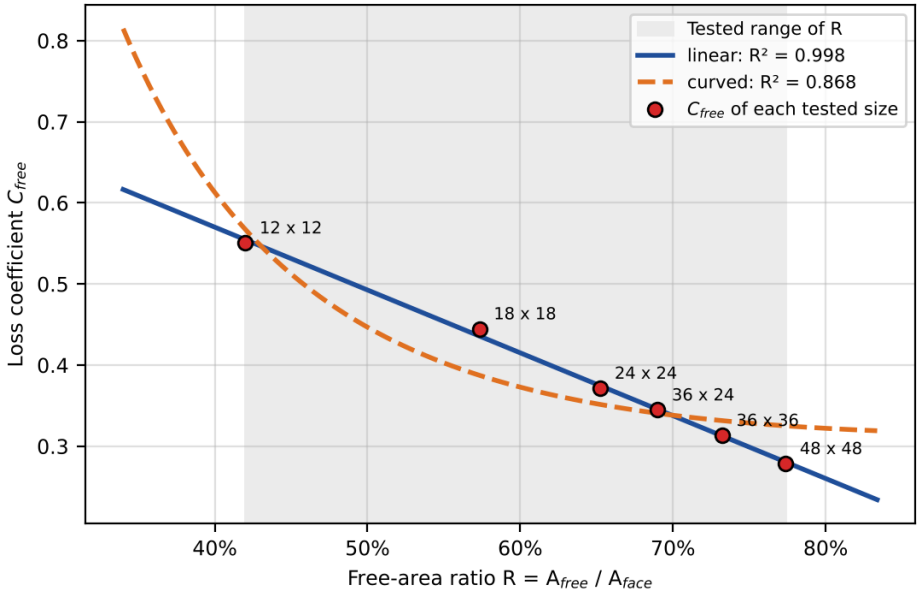


Figure 3. Both size-model shapes for SAMPLE-FSD-100. Each red point is one tested size's fitted C_{free}. Outside the shaded tested range the two shapes diverge; real data usually curves like the dashed line at small R.

5d. Blending next to a tested size

A tested size uses its own fitted coefficient, which is usually a little above or below the size model. Switching straight from one to the other would make the pressure drop jump between a tested size and a size a fraction of an inch away. Instead, an untested size within 15% of a tested size, in both width and height, keeps part of that size's offset from the size model:

$$C_{free} = C_{model}(R) + w \times [\, C_{tested} - C_{model}(R_{tested}) \,]$$

$$w = 1 - d \,/\, \ln(1 + 0.15), \quad d = \max(\, |\ln(W \,/\, W_{tested})|,\, |\ln(H \,/\, H_{tested})| \,), \quad w \geq 0$$

w is 1 at the tested size and falls linearly to 0 at 15% away; the nearest tested size (largest w) is used. Keeping the offset, rather than the tested coefficient itself, means the size model still accounts for the difference in free-area ratio. The Estimate tab shows the size-model value, the nearest tested size, its offset and the weight. A free area typed in as an override uses the size model at that ratio, with no blending.

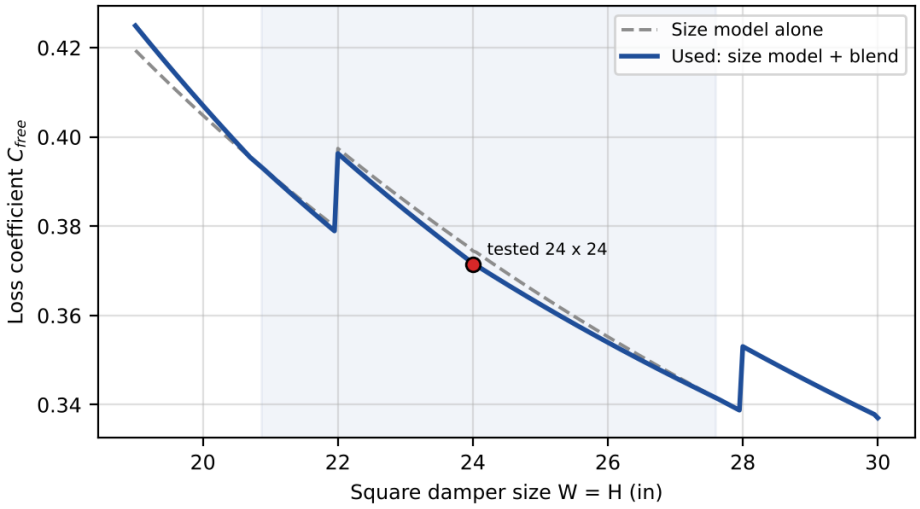


Figure 4. C_{free} for square sizes around the tested 24 x 24 (SAMPLE-FSD-100 with its geometry calibrated to the measured free areas). Shaded: the ±15% blend zone. The curve passes smoothly through the tested size; the steps at 22 and 28 in are the blade count changing.

5e. Which coefficient is used

Situation	Coefficient used
The exact size was tested (within 0.01 in)	That size's own fitted C_{free}
Single-section size within 15% of a tested size	Size model at the damper's R, plus the weighted offset of the nearest tested size (5d)
Multi-section damper	Size model at the R of the whole assembly, joints included, plus the weighted offset of the tested size nearest the **section** size. If the section size was tested and there is no joint blockage, this is exactly that section's C_{free}: every section sees the same face velocity, so it behaves like the tested damper.
Any other size	Size model at the damper's free-area ratio R
Free area override entered	Size model at the ratio given by the override

For a multi-section damper the size model is read at the ratio of the whole assembly, **including** the joint blockage. The joints block each section like extra frame, so they reduce its effective free-area ratio and raise its coefficient as well as the velocity. (Program versions before 0.5 read the size model at the section's ratio without the joints, which gave a slightly lower, less conservative coefficient.)

Catalogs often quote a coefficient based on face velocity instead. The two are related by $C_{face} = C_{free} / R^2$; the program shows both.

7. Step 6: pressure drop

$$\Delta P = C_{free} \times VP_{free} \quad \text{(in. wg)} \qquad \Delta P \text{ (Pa)} = \Delta P \text{ (in. wg)} \times 249.089$$

The curve drawn on the Estimate tab repeats this calculation over a range of airflows for the same damper and air conditions.

8. Worked example 1: a tested single-section size

SAMPLE-FSD-100, 24 in × 24 in, 4,000 cfm, standard air (70 °F, sea level).

Step	Calculation	Result
1. Size and sections	Within limits (width 8 to 144 in, height 8 to 96 in; max section 48 x 48 in, 0.5 in extra at joints). 24 ≤ 48, so 1 section.	1 section, 24 × 24
2. Free area	24 × 24 is in the measured free-area table, so it takes priority. (The geometry estimate would be 399 in² = 2.771 ft², with 4 blades from the blade table row 22-27.99 in.)	A_{free} = 2.610 ft²
	A_{face} = 24 × 24 / 144 = 4.000 ft²; R = 2.610 / 4.000	R = 0.6525 (65.2%)
3. Velocities	V_{face} = 4,000 / 4.000; V_{free} = 4,000 / 2.610	1,000 fpm; 1,532.6 fpm
4. Velocity pressure	ρ = 0.075 (standard air); VP = (1,532.6 / 4005)²	VP_{free} = 0.14643 in. wg
5. Coefficient	24 × 24 was tested, so its own fit is used (table below)	C_{free} = 0.3715
6. Pressure drop	0.3715 × 0.14643	**ΔP = 0.0544 in. wg** (13.5 Pa)

Where C_{free} = 0.3715 comes from

The test data for 24 × 24 (free area 2.610 ft², so $V_{free} = V_{face} \times 4.000 / 2.610$):

V_{face} (fpm)	V_{free} (fpm)	x = VP_{free} (in. wg)	y = measured ΔP	x·y	x²	C·x (fitted)	Residual
500	766.3	0.03661	0.014	0.000513	0.001340	0.0136	+2.9%
1,000	1,532.6	0.14643	0.054	0.007907	0.021442	0.0544	-0.7%
1,500	2,298.9	0.32947	0.124	0.040854	0.108551	0.1224	+1.3%
2,000	3,065.1	0.58573	0.226	0.132374	0.343074	0.2176	+3.9%
2,500	3,831.4	0.91520	0.332	0.303845	0.837583	0.3400	-2.3%
3,000	4,597.7	1.31788	0.491	0.647080	1.736812	0.4896	+0.3%
			Sums	**1.132573**	**3.048802**		

$$C_{free} = \Sigma xy / \Sigma x^2 = 1.132573 / 3.048802 = 0.3715 \qquad n = 1.99$$

The same damper in hot air at altitude

At 120 °F and 5,000 ft, $\rho = 0.075 \times (1 - 6.8754\times10^{-6} \times 5000)^{5.2559} \times 529.67 / 579.67 = 0.05702$ lb/ft³. Only the velocity pressure changes, by the factor ρ / 0.075 = 0.7603, so ΔP = 0.0544 × 0.7603 = **0.0414 in. wg**.

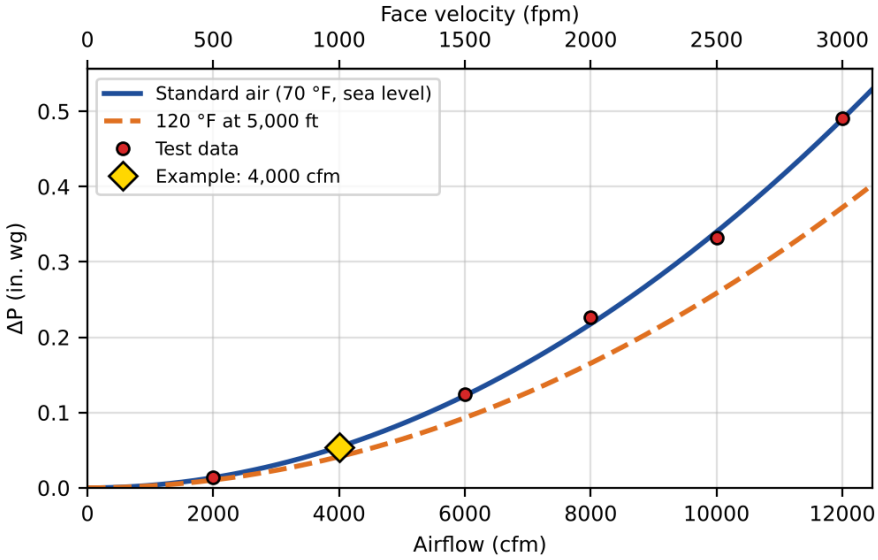


Figure 5. The resulting curve for the 24 x 24 damper, with its test data and the example operating point. Lighter air shifts the whole curve down.

9. Worked example 2: a size next to a tested one

SAMPLE-FSD-100, 26 in × 26 in, 4,000 cfm, standard air. 26 × 26 was not tested and is not in the measured free-area table, but it is within 15% of the tested 24 × 24.

Step	Calculation	Result
1–2. Sections, free area	1 section. Geometry estimate: 4 blades (blade table), 483.0 in²; A_{face} = 4.694 ft²	A_{free} = 3.354 ft², R = 0.7145
3–4. Velocity pressure	V_{free} = 4,000 / 3.354; VP = (1,192.5 / 4005)²	VP_{free} = 0.08866 in. wg
5a. Size model	0.8799 − 0.7747 × 0.7145	C_{model} = 0.3264
5b. Offset of 24 × 24	0.3715 − (0.8799 − 0.7747 × 0.6525)	-0.0030
5c. Weight	d = ln(26 / 24) = 0.0800; w = 1 − 0.0800 / ln(1.15)	w = 0.427
5d. Coefficient	0.3264 + 0.427 × (-0.0030)	C_{free} = 0.3252
6. Pressure drop	0.3252 × 0.08866	**ΔP = 0.0288 in. wg** (7.2 Pa)

Here the 24 × 24 offset is small (-0.0030), so blending changes C_{free} by only 0.0013; with real, scattered test data the offsets, and the effect of blending, are larger. Note that 26 × 26 takes its free area from the uncalibrated sample geometry (R = 71.4%), while 24 × 24 uses its measured 65.2%. Once the geometry is calibrated, the free-area ratios of neighbouring sizes line up as well (Figure 4).

10. Worked example 3: an untested multi-section size

SAMPLE-FSD-100, 96 in × 72 in, 40,000 cfm, standard air. The model's size rules allow up to 144 × 96 in overall, with sections of at most 48 × 48 in and 0.5 in of extra blockage at each joint.

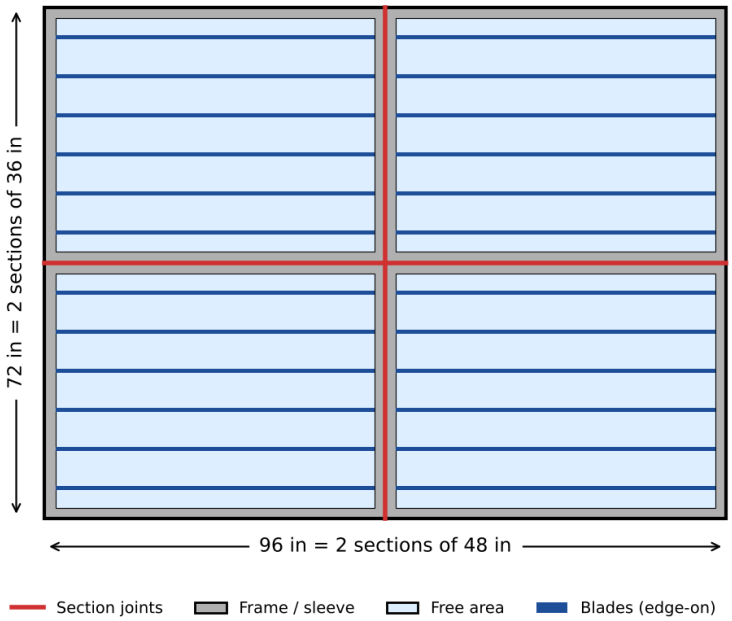


Figure 6. The 96 x 72 damper is split into 2 x 2 equal sections of 48 x 36 in, each with its own frame and 6 blades.

Step	Calculation	Result
1. Size and sections	96 ≤ 144 and 72 ≤ 96, within limits. n_w = ⌈96 / 48⌉ = 2, n_h = ⌈72 / 48⌉ = 2	4 sections of 48 × 36
2. Blades	Section height 36 in is in the blade-table row 34-39.99 in	N = 6 per section
Free area per section	w_{open} = 48 − 2 × 1.5 = 45; h_{open} = 36 − 2 × 1.5 = 33; 45 × 33 − 6 × 0.5 × 45 − 0	$A_{section}$ = 1,350 in²
Joints	0.5 × [(2 − 1) × 72 + (2 − 1) × 96] − 0.5² × 1 × 1	A_{joints} = 83.75 in²
Total free area	4 × 1,350 − 83.75 = 5,316.25 in² (not in the measured table, so the geometry estimate is used)	A_{free} = 36.918 ft²
	A_{face} = 96 × 72 / 144 = 48.000 ft²	R = 0.7691 (76.9%)
3. Velocities	V_{face} = 40,000 / 48; V_{free} = 40,000 / 36.918	833.3 fpm; 1,083.5 fpm
4. Velocity pressure	VP = (1,083.5 / 4005)²	VP_{free} = 0.07319 in. wg
5. Coefficient	Neither 96 × 72 nor a 48 × 36 section was tested, and no tested size is within 15% of the section size, so there is no blend. Size model at the assembly ratio, joints included: 0.8799 − 0.7747 × 0.7691	C_{free} = 0.2841
6. Pressure drop	0.2841 × 0.07319	**ΔP = 0.0208 in. wg** (5.2 Pa)

Change from version 0.4. Earlier versions read the size model at the section ratio without the joints, R = 0.7691 + 83.75 / 6,912 = 0.7812, giving C_{free} = 0.2747 and ΔP = 0.0201 in. wg. Including the joints raises the estimate by 3.4%. The assembly ratio of 76.9% is inside the tested range (42.0% to 77.4%), so no extrapolation warning is shown.

11. Warnings and errors

An estimate is still given, with a warning, when:

Warning	Meaning
Free-area ratio outside the tested range	The size model is extrapolated beyond the ratios it was fitted to.
Face area (or section face area) outside the tested sizes	The size is smaller or larger than anything tested.
Face velocity above the highest tested velocity (by more than 5%)	The velocity-squared law is extrapolated.
Face velocity below the lowest tested velocity (by more than 5%)	Loss coefficients tend to rise at low velocity (low Reynolds number), so the real pressure drop may be somewhat higher than the estimate.
Test data follows V^n with n outside 2 ± 0.2	The tested size used for the coefficient does not follow the velocity-squared law well; check its test points.
Fewer than 3 test points for this size	The size's own fit is not well constrained.
Section height outside the blade table	The blade count came from the blade pitch instead.

The two extrapolation warnings on ratio and face area are not shown when the coefficient comes entirely from a tested size (weight 1). No estimate is given, and the reason is shown, when the size is outside the model's limits, a section height falls in a gap in the blade table, the frame or blades leave no opening, the free area is not smaller than the face area, or the size model gives a coefficient of zero or less.

12. Assumptions and limitations

• **Fully open blades.** The damper is modeled fully open, which is normal operation for fire/smoke dampers. Partly closed blades are not modeled.

• **Velocity-squared law.** Pressure drop is taken as proportional to velocity squared, which holds for the turbulent flow found in ducts. The fitted exponent n (5b) shows how well each tested size follows it.

• **Test configuration.** Pressure drop depends on how the damper was installed in the test (AMCA 500-D configuration). Estimates apply to the configuration the test data came from, and comparisons between models are only like-for-like when their data comes from the same configuration. Published data for UL 555S fire/smoke dampers is often figure 5.3 (ducted both sides). System effects from nearby elbows, transitions or other fittings are not included.

• **Standard-air test data.** Test pressure drops are assumed to be corrected to standard air. The density correction covers temperature and altitude, not humidity.

• **Simple geometry.** The geometry estimate assumes a rectangular frame, horizontal blades of constant projected thickness, equal sections, and a fixed other blockage per section. Calibrating to measured free areas absorbs most of the difference for real dampers.

• **Size model.** Between tested sizes, C_{free} is assumed to follow the chosen shape (straight line or curve) in the free-area ratio. Estimates are most reliable inside the tested range of sizes and velocities; outside it, the curved shape is usually the safer of the two at small ratios.

Appendix: symbols and units

Symbol	Meaning	Unit
W, H	Nominal damper width and height (overall)	in
W_s, H_s	Section width and height	in
n_w, n_h	Number of sections across the width and up the height	–
f_w, f_h	Frame obstruction on each side (left/right, top/bottom)	in
t	Blade thickness facing the airflow	in
p	Blade pitch (used only outside the blade table, on the open height)	in
N	Blades per section	–
O	Other blockage per section	in^2
j	Extra blockage width along each section joint	in
A_{face}, A_{free}	Face area and free area	ft^2
R	Free-area ratio A_{free} / A_{face}	–
Q	Airflow	cfm
V_{face}, V_{free}	Face velocity and velocity through the free area	fpm
ρ	Air density (0.075 for standard air)	lb/ft^3
VP	Velocity pressure	in. wg
C_{free}, C_{face}	Loss coefficient based on free-area / face velocity	–
C_{model}	Size-model coefficient at a given R	–
a, b	Fitted size-model constants	–
w	Blend weight towards the nearest tested size (0 to 1)	–
n	Fitted exponent of $\Delta P = k \times V^n$	–
ΔP	Pressure drop	in. wg